21 Conducting putty is a substance that conducts electricity and can be molded into various shapes. A cylinder made from this putty has a length L and cross-sectional area A. A potential difference V is applied across its two ends. The drift velocity of the charge carriers in the putty is v_d.

This cylinder is then reshaped to one with a cross-sectional area $\frac{A}{4}$. The same potential difference V is again applied across its ends.

What is the new drift velocity of the charge carriers?

A $\frac{v_d}{4}$ **B** $\frac{v_d}{2}$ **C** v_d **D** $4v_d$